AI-Powered Research Gap Identifier & Trend Analyzer

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Academic research supervision requires continuous monitoring of student progress, effective communication and timely feedback. However, supervisors managing multiple research groups often face difficulties reviewing lengthy research logbooks, tracking research activities and maintaining consistent communication with students. Existing academic supervision systems primarily support administrative tasks and progress tracking, with limited capabilities for automatically interpreting and summarising unstructured research logbooks. To address these challenges, this research presents a Collaborative Workspace and NLP-Powered Supervisor Assistant integrated into ResearchLens, an academic research supervision platform.

The proposed solution combines a structured collaborative workspace with an automated logbook summarisation mechanism. The workspace supports research group management, student logbook submissions, supervisor review and feedback, real-time communication, task management and development activity monitoring. The NLP component employs a fine-tuned Bidirectional and Auto-Regressive Transformers (BART) model to generate concise summaries of student research activities. The model was fine-tuned using 100 academic research logbook samples and evaluated using ROUGE-1, ROUGE-2 and ROUGE-L metrics. Comparative experiments included extractive summarisation techniques, pretrained BART, fine-tuned Transformer models and zero-shot ChatGPT.

The fine-tuned BART model achieved ROUGE-1, ROUGE-2 and ROUGE-L scores of 0.4785, 0.1842 and 0.3810, respectively, outperforming the evaluated alternatives in lexical overlap with reference summaries. The developed system integrates automated summaries into the supervisor dashboard while preserving access to original logbooks for detailed verification and feedback. The individual contribution includes developing the collaborative workspace, implementing the logbook management and supervisor review workflows, and integrating the NLP summarisation service. The findings demonstrate the potential of combining collaborative research management with automated summarisation to support academic supervision while maintaining human oversight.

Keywords: Academic Research Supervision, Collaborative Workspace, Natural Language Processing, BART, Abstractive Summarisation

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
BART	Bidirectional and Auto-Regressive Transformers
BERT	Bidirectional Encoder Representations from Transformers
CRUD	Create, Read, Update and Delete
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JWT	JSON Web Token
LLM	Large Language Model
NLP	Portable Document Format
REST	Representational State Transfer
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
T5	Text-to-Text Transfer Transformer

1 INTRODUCTION

1.1 Overall Project Context

Academic research is an essential part of undergraduate education, where students are required to apply theoretical knowledge to practical research problems. The research process involves identifying problems, reviewing literature, developing solutions, conducting experiments and documenting progress. Throughout this process, effective communication between students and supervisors is important for maintaining research quality and achieving the intended objectives. Regular progress monitoring and constructive feedback help students identify challenges and improve their research activities [1].

Academic supervisors are often responsible for guiding multiple student research groups simultaneously. This involves reviewing research documentation, monitoring progress, evaluating completed activities and providing timely feedback. Research logbooks are commonly used to document students' progress, including completed tasks, technical implementations, challenges and planned activities. However, reviewing lengthy logbooks from several research groups can be time-consuming, particularly when the information is presented as detailed, unstructured text.

Existing academic project management systems provide facilities for organising research information, managing student groups and monitoring project activities [2], [6]. Learning analytics dashboards also support academic monitoring by presenting structured information such as activity records and progress indicators [3]. Although these approaches support research administration and progress tracking, they provide limited assistance in interpreting the detailed narrative information contained in student research logbooks.

Advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP) provide opportunities to address this limitation. Transformer-based models, particularly Bidirectional and Auto-Regressive Transformers (BART), support abstractive text summarisation by generating concise representations of longer documents [4]. Applying such models to academic research logbooks can help

supervisors obtain an initial overview of student activities without immediately reviewing every detail of the original submission.

Within this context, ResearchLens, an AI-Powered Research Gap Identifier and Trend Analyzer, is developed as an academic research support platform. The Collaborative Workspace & NLP-Powered Supervisor Assistant is an individual research component within this wider project. It combines structured student and supervisor workflows with automated research logbook summarisation.

The collaborative workspace supports research logbook submissions, supervisor review, feedback management and research collaboration. The NLP component extends this functionality by using a fine-tuned BART model to generate concise summaries of submitted research activities. These summaries are presented through the supervisor dashboard alongside the original logbooks, allowing supervisors to examine the information before providing feedback.

By connecting collaborative research management with automated summarisation, the proposed component aims to support a more organised academic supervision process. Importantly, the system is intended to assist supervisors rather than replace their academic judgement. Supervisors remain responsible for verifying research information, evaluating student progress and providing final feedback.

1.2 Individual Component Context

The individual research component, titled **Collaborative Workspace & NLP-Powered Supervisor Assistant**, forms part of the wider ResearchLens academic research support platform. It focuses on improving the management of student research activities and supporting supervisors in reviewing research progress. The component combines two interconnected functions: a collaborative workspace for student and supervisor interactions and a Natural Language Processing (NLP) mechanism for automated research logbook summarisation.

The collaborative workspace provides a centralised environment for managing academic research activities. The workspace also provides facilities for viewing research records, communicating with supervisors and accessing feedback. These

functions support a structured research supervision process and reduce the need to manage related activities through separate applications.

A major part of the individual development contribution is the research logbook management system. The student interface allows research activities to be recorded with automatic entry numbering and duplicate submission validation. On the supervisor side, the developed dashboard provides access to research groups, submitted logbooks and feedback management. These functions connect student progress documentation with the supervisor review process.

The workspace also incorporates real-time communication using Socket.IO, GitHub repository analytics, Microsoft Planner integration and supervisor announcements. These supporting functions allow students and supervisors to communicate, manage research tasks and access software development activity information within the research environment.

The second major contribution focuses on automated research logbook summarisation using a fine-tuned Bidirectional and Auto-Regressive Transformers (BART) model. BART is a sequence-to-sequence Transformer architecture capable of generating concise textual representations of longer documents [4]. In this research, the model is adapted to summarise academic research logbooks containing descriptions of completed activities, technical challenges and research progress.

The NLP component is implemented as a Python-based FastAPI microservice connected to the main Node.js and Express backend. When research logbook content is processed, the microservice generates a summary using the fine-tuned BART model. The generated summary is subsequently made available through the supervisor review interface alongside the original logbook. The application uses React for the frontend and Supabase PostgreSQL for persistent data storage.

The component additionally incorporates an AI-assisted feedback suggestion mechanism using Google Gemini. This mechanism generates contextual suggestions based on summarised research activities, which supervisors can review and modify before finalising their feedback. Automated summaries and feedback suggestions

therefore function as supporting information rather than independent academic evaluations.

1.3 Problem Statement

Academic research supervisors managing multiple student research groups face challenges in efficiently monitoring research progress, reviewing lengthy and unstructured research logbooks, and providing timely, constructive feedback due to fragmented research management and communication processes and the limited integration of automated logbook summarisation within existing academic supervision platforms.

Research logbooks contain important information about completed activities, technical challenges and research progress. However, supervisors must manually examine these records to identify relevant information, which can increase their review workload. Although existing academic supervision systems support project management, communication and progress monitoring, they provide limited support for automatically summarising unstructured research documentation within a unified supervision workflow [2], [3], [6].

1.4 Research Gap

Existing academic research supervision platforms provide facilities for managing research projects, organising student activities, maintaining documentation and supporting communication between students and supervisors [2], [6], [8]. Learning analytics dashboards further support progress monitoring by analysing structured information, such as participation records, completed tasks and activity indicators [3], [9]. However, these approaches primarily focus on administrative management and quantitative progress tracking, with limited support for interpreting the unstructured textual information contained in academic research logbooks.

Research logbooks often contain detailed descriptions of completed activities, technical challenges, methodological decisions and planned work. Although automated text summarisation offers a potential solution, general-purpose summarisation models are not specifically adapted to the structure and terminology of academic research documentation. Transformer-based models such as BART, T5

and PEGASUS have demonstrated capabilities in abstractive summarisation [4], [12], [14]. Nevertheless, their application to academic research logbooks requires domain-specific adaptation and evaluation to determine their effectiveness in representing research progress.

Therefore, the research gap addressed by this study is the limited integration and evaluation of domain-specific automated research logbook summarisation within collaborative academic supervision platforms.

1.5 Aim and Objectives

The aim of the individual component is to design, develop and evaluate an integrated collaborative academic research supervision platform that combines structured student and supervisor workflows with a fine-tuned BART-based NLP summarisation model to support research progress monitoring, automated logbook summarisation and constructive supervisor feedback while maintaining human oversight.

Table 1.1: Individual Research Objectives and Where Each Is Addressed

No.	Individual Objective	Addressed In
O1	Design and implement a collaborative academic research supervision workspace supporting student logbook submissions, supervisor review, progress monitoring and feedback management.	Sections 4.1, 4.4, 5.2
O2	Integrate the fine-tuned BART summarisation service and AI-assisted feedback suggestions into the supervisor review workflow while maintaining human oversight.	Sections 4.2, 5.3, 6.4
O3	Implement real-time communication, GitHub analytics, Microsoft Planner integration, supervisor announcements and notifications to support research collaboration.	Sections 4.4, 5.2, 6.2
O4	Implement and verify research logbook management functions, including automatic entry numbering, duplicate submission prevention, chronological activity	Sections 5.2, 5.4, 6.2

	management and institutional print formatting.	

1.6 Research Questions

RQ1: How effectively can a domain-specific fine-tuned BART model summarise academic research logbooks compared with existing extractive and abstractive summarisation approaches?

RQ2: How can a collaborative workspace be designed and implemented to support research progress management, communication and feedback between students and academic supervisors?

RQ3: How can automated logbook summarisation and AI-assisted feedback suggestions be integrated into the supervisor review workflow while maintaining human oversight?

RQ4: How can the functionality and integration of the collaborative workspace and NLP summarisation component be verified against the intended academic supervision workflow?

1.7 Scope Limitations and Assumptions

This research focuses on developing a collaborative academic supervision workspace integrated with a fine-tuned BART model for automated research logbook summarisation. The workspace supports logbook submissions, supervisor review, feedback, communication and research progress monitoring. The NLP component uses 100 academic research logbooks for model training and evaluation. The study is limited by the relatively small dataset and the use of ROUGE metrics, which measure lexical overlap rather than factual accuracy or practical usefulness. Formal supervisor evaluations and quantitative measurements of supervision efficiency were not conducted.

The system assumes that students provide sufficient textual information in their logbooks and that users have appropriate access to the platform and its integrated services.

1.8 Individual Contribution

The individual contribution focuses on developing the Collaborative Workspace & NLP-Powered Supervisor Assistant within the ResearchLens platform. Developed the student logbook submission and supervisor review interfaces, including automatic entry numbering, duplicate submission prevention, chronological activity management and feedback handling. Additional contributions include real-time communication using Socket.IO, GitHub analytics, Microsoft Planner integration, supervisor announcements and notifications. The NLP contribution involved preparing academic research logbook data, fine-tuning and evaluating the BART summarisation model, and implementing a FastAPI microservice for automated summary generation. Also integrated Google Gemini to provide AI-assisted feedback suggestions that supervisors can review and modify. These components were connected to create a unified supervision workflow in which students submit research activities, the NLP service generates summaries and supervisors review the information before providing final feedback.

1.9 Thesis Structure

This thesis consists of nine chapters. Chapter 1 introduces the research, while Chapter 2 reviews related literature. Chapter 3 explains the research methodology, and Chapter 4 presents the system design. Chapter 5 describes implementation, integration and testing. Chapter 6 presents the results and validation, followed by the discussion in Chapter 7. Chapter 8 addresses commercialisation and sustainability, while Chapter 9 concludes the research and outlines future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

This literature review examines existing research on academic supervision, collaborative research platforms, student progress monitoring and Natural Language Processing (NLP) based text summarisation.

The review focuses on three main areas. The first examines technology-supported academic supervision, including communication, research management and feedback mechanisms [1], [2], [8]. The second considers learning analytics and software development monitoring techniques used to track student activities and project progress [3], [9], [10], [11]. The third explores automated summarisation methods, particularly Transformer-based models such as BART, T5 and PEGASUS, alongside general-purpose Large Language Models [4], [5], [12], [14].

2.2 Domain and Theoretical Foundations

2.2.1 Academic research supervision and collaboration

Academic research supervision involves continuous communication, progress monitoring and constructive feedback between students and supervisors. Effective feedback helps students identify weaknesses, overcome research challenges and improve their academic work [1]. Technology-supported supervision platforms facilitate research coordination, document sharing and communication, allowing students and supervisors to manage research-related information within digital environments [2], [8].

Collaborative research platforms extend these capabilities by providing shared workspaces for managing research activities and maintaining supervision records. However, managing research information alone does not address the difficulty of interpreting lengthy textual progress reports. This distinction supports the integration of collaborative research management with automated logbook summarisation.

2.2.2 Research progress monitoring

Learning analytics systems analyse student activity records and progress indicators to support educational monitoring [3], [9]. In education, Git repository information can

also provide evidence of development activities and individual contributions [10], [11].

Although these approaches provide useful quantitative information, they cannot fully explain the research decisions, challenges and methodological progress recorded in narrative logbooks. Therefore, textual analysis can complement structured progress monitoring within an academic supervision platform.

2.2.3 NLP and automated text summarisation

Automated text summarisation aims to produce a shorter representation of a document while retaining its important information. Two main approaches are extractive and abstractive summarisation.

Extractive techniques, such as Lead-3 and LexRank, select existing sentences from the source document. In contrast, abstractive techniques generate new sentences representing the main information. Transformer-based models, including BART, T5 and PEGASUS, support abstractive summarisation through sequence-to-sequence learning [4], [12], [14].

BART combines bidirectional encoding with autoregressive decoding and uses denoising pretraining to support natural language generation [4]. Its architecture allows a pretrained model to be fine-tuned using domain-specific data. This makes it relevant to academic research logbooks, which contain chronological descriptions of technical activities and research progress.

2.3 Related Research

2.3.1 Technology-supported academic supervision

Previous research highlights the importance of constructive feedback in supporting students' academic development and research progress [1]. Information and communication technologies have also been investigated for supporting academic research activities, document management and interactions between students and supervisors [2]. Similarly, technology-enabled supervision approaches demonstrate how digital environments can facilitate communication and research coordination [8].

Although these studies establish the importance of digital academic supervision, they provide limited evidence of integrating automated research logbook summarisation directly into structured supervision workflows.

2.3.2 Academic project management and progress monitoring

Academic project management systems have been developed to organise project information and support administrative processes [6]. Monitoring systems for project-based learning also provide mechanisms for tracking student activities and task progress [7].

Learning analytics dashboards extend these capabilities by presenting structured educational information, including participation records and activity indicators, to support progress monitoring [3], [9]. In software engineering education, Git logs and user stories have been investigated for assessing individual development contributions [10]. Process mining techniques have also been applied to analyse Git repository activities and development patterns [11].

However, these approaches primarily focus on structured information rather than interpreting detailed narrative research documentation. Consequently, they may not fully represent the technical challenges, methodological decisions and research progress described in student logbooks.

2.3.3 Transformer-based text summarisation

Transformer-based models have introduced advanced approaches to abstractive text summarisation. BART uses denoising pretraining with a sequence-to-sequence architecture to support natural language generation [4]. T5 applies a unified text-to-text framework to NLP tasks, including summarisation [14]. PEGASUS uses a summarisation-oriented pretraining approach to generate concise representations of source documents [12].

Although these architectures support abstractive summarisation, their effectiveness depends on the characteristics of the target documents. Academic research logbooks contain chronological activity descriptions and technical terminology that may require domain-specific model adaptation.

General-purpose Large Language Models, including ChatGPT, also provide text generation and summarisation capabilities. However, their application in educational environments introduces considerations related to reliability, data privacy and responsible use [5].

2.4 Existing Systems Products and Techniques.

Existing academic supervision and project management systems provide facilities for managing research activities, maintaining documentation and supporting student and supervisor communication [2], [6]–[8]. Learning analytics dashboards and Git-based monitoring techniques extend these capabilities by providing structured information about student participation and software development activities [3], [9]–[11]. However, these approaches provide limited support for interpreting the narrative content of academic research logbooks.

Automated text summarisation techniques offer an alternative for processing unstructured documentation. Extractive approaches, such as Lead-3 and LexRank, select existing sentences, while Transformer-based abstractive models, including BART, T5 and PEGASUS, generate condensed textual representations [4], [12], [14]. General-purpose LLMs also provide summarisation capabilities, although their application introduces considerations related to output consistency and data privacy [5].

Table 2.1: Comparison of existing academic supervision and automated summarisation approaches

Source	Primary focus	Relevance	Limitations
Stein and Sim [2]; Sim et al. [8]	Technology-supported academic supervision	Supports student and supervisor communication, research coordination and feedback	Limited integration of automated research logbook summarisation
Arumugam et al. [6]; Desyatirikova et al. [7]	Academic project management and progress monitoring	Provides structured management of research activities and project information	Limited interpretation of unstructured research documentation
Masiello et al. [3]; Sušnjak et al. [9]	Learning analytics dashboardsmonitoring	Supports monitoring student activities using structured	Cannot fully represent narrative research progress

		progress indicators	
Buffardi [10]; Mačák et al. [11]	Git-based software development monitoring	Provides evidence of software development activities and individual contributions	Does not fully capture research reasoning and methodological challenges
Lewis et al. [4]	BART-based abstractive summarisation	Provides the Transformer architecture used for domain-specific logbook summarisation	Pretrained models may require adaptation to academic research documentation
Zhang et al. [12]; Raffel et al. [14]	PEGASUS and T5 summarisation models	Provides alternative Transformer models for comparative evaluation	Performance depends on domain-specific data and evaluation conditions
Kasneci et al. [5]	Large Language Models in education	Provides context for using general-purpose LLMs in academic applications	Reliability, data privacy and responsible use considerations

2.5 Gap Synthesis and Conceptual Framework

The reviewed literature identifies three areas relevant to the proposed research: technology-supported academic supervision, research progress monitoring and automated text summarisation. Although existing approaches support these functions individually, limited attention has been given to integrating domain-specific research logbook summarisation into a collaborative supervision workflow.

Academic project management and learning analytics systems primarily support structured activity tracking, while Git-based monitoring provides information about software development contributions [3], [6], [9]–[11]. These approaches offer limited insight into the research challenges and methodological decisions documented in narrative logbooks. Meanwhile, Transformer-based models provide capabilities for abstractive summarisation, but their application to academic research documentation requires domain-specific adaptation [4], [12], [14].

The proposed Collaborative Workspace & NLP-Powered Supervisor Assistant addresses this gap through an integrated approach. Students maintain research logbooks within a shared workspace, while a fine-tuned BART model generates concise summaries to support supervisor review. The workspace also incorporates communication, progress monitoring and AI-assisted feedback suggestions, with final feedback decisions remaining under human supervision.

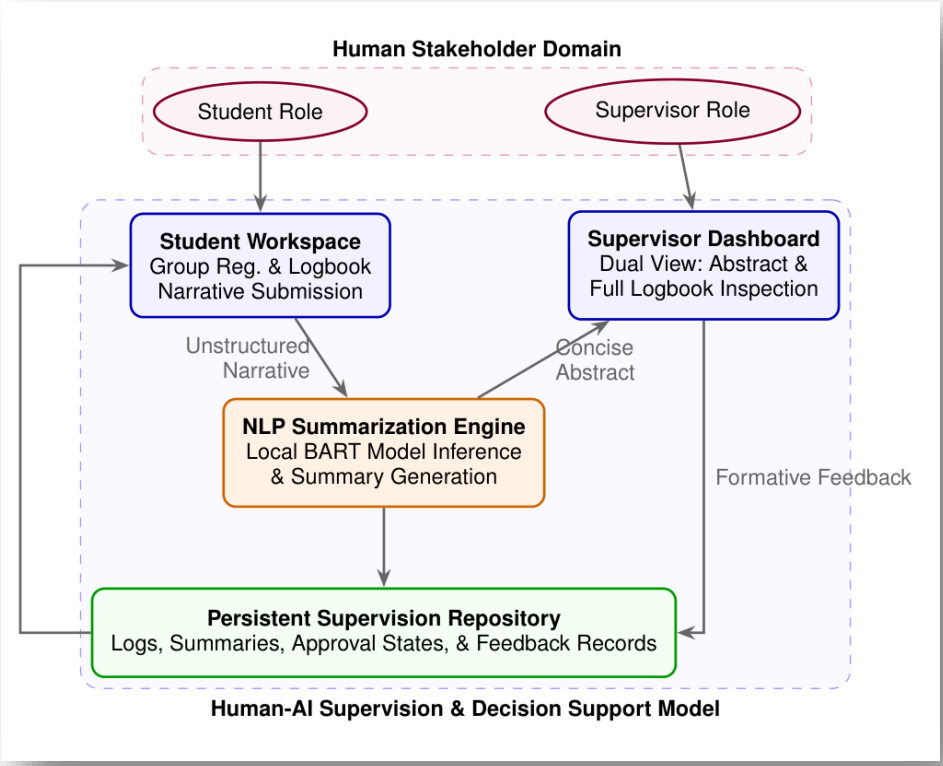


Figure 2.1: Conceptual Framework of the Collaborative Workspace and NLP-Powered Supervisor Assistant

Figure 2.1 illustrates the relationship between research logbook management, NLP-based summarisation and supervisor review within the collaborative workspace. This framework provides the basis for the research methodology and system design presented in the following chapters.

3 RESEARCH METHODOLOGY

3.1 Research Design

The research consists of three main stages. First, a BART-based summarisation model is fine-tuned using academic research logbooks and corresponding reference summaries. The model's summarisation performance is evaluated using ROUGE-1, ROUGE-2 and ROUGE-L metrics. Its performance is also compared with extractive and abstractive summarisation baselines to assess its effectiveness.

Second, the fine-tuned model is integrated into the Collaborative Workspace & NLP-Powered Supervisor Assistant through an NLP microservice. The workspace supports student logbook submission, supervisor review, AI-assisted feedback suggestions, communication and research progress monitoring.

Third, the implemented components are examined through functional and integration testing to verify their intended behaviour and interactions. This includes checking logbook management, summary generation, supervisor feedback and communication workflows.

3.2 Data, Participants and Study Materials

The primary dataset used in this research consists of 100 academic research logbooks and their corresponding reference summaries. These materials were used to fine-tune and evaluate the BART-based summarisation model. The dataset was divided into training, validation and testing subsets, as presented in Table 3.1.

Table 3.1: Distribution of Academic Research Logbook Dataset

Dataset Split	Percentage (%)	Number of Samples	Purpose
Training Set	80%	80	Used to fine-tune the pretrained BART model for academic research logbook summarisation.
Validation Set	10%	10	Used to monitor model performance during training.
Testing Set	10%	10	Used to evaluate the fine-tuned model using ROUGE metrics.
Total	100%	100	Complete academic research logbook dataset.

The research logbooks were collected from students and used as the primary study materials. These logbooks contained records of students' research activities, completed tasks, progress updates and challenges encountered during their academic projects. The collected logbooks formed the basis for preparing the dataset used to fine-tune and evaluate the BART summarisation model.

3.3 Experimental and Study Procedure

3.3.1 Data preparation and model fine-tuning

The collected academic research logbooks and their corresponding reference summaries were organised into training, validation and testing subsets. The logbook texts were preprocessed and tokenised using the BART tokenizer to prepare them for model training.

The pretrained facebook/bart-large-cnn model was fine-tuned using the training dataset. The training configuration included a maximum input length of 512 tokens, a maximum target length of 64 tokens, a learning rate of 3×10^{-5}, a batch size of 2 and 10 training epochs. Mixed-precision training (FP16) was used during fine-tuning.

3.3.2 Model evaluation

Following fine-tuning, the model generated summaries for the testing dataset using beam search with four beams. The generated summaries were compared with their reference summaries using ROUGE-1, ROUGE-2 and ROUGE-L.

3.3.3 System integration and verification

The fine-tuned model was deployed through a FastAPI-based NLP microservice and connected to the collaborative workspace. Within the implemented workflow, students submit research logbooks, and supervisors access generated summaries alongside the original entries.

The workspace also incorporates AI-assisted feedback suggestions, real-time communication, GitHub repository analytics and Microsoft Planner integration. The generated feedback suggestions remain subject to supervisor review and editing.

3.4 Measures and Evaluation Metrics

The summarisation model was evaluated using three ROUGE metrics:

- **ROUGE-1:** Measures unigram overlap between generated and reference summaries.
- **ROUGE-2:** Measures bigram overlap, indicating how well consecutive word sequences are preserved.
- **ROUGE-L:** Measures the longest common subsequence between generated and reference summaries, reflecting similarities in word sequence.

These metrics were used to compare the fine-tuned BART model against Lead-3, LexRank, pretrained BART, T5, PEGASUS and ChatGPT.

The technical evaluation considered whether the implemented components supported their intended operations, including:

1. Research logbook submission, entry numbering and duplicate prevention.
2. NLP microservice communication and automated summary generation.
3. Supervisor review and editing of AI-assisted feedback suggestions.
4. Real-time communication and integration with GitHub analytics and Microsoft Planner.

3.5 Data Analysis

The generated summaries were compared with their corresponding reference summaries using ROUGE-1, ROUGE-2 and ROUGE-L. The resulting scores were used to assess the model's ability to retain important textual information from academic research logbooks.

A comparative analysis was conducted using six baseline approaches: Lead-3, LexRank, pretrained BART, fine-tuned T5, fine-tuned PEGASUS and ChatGPT. The ROUGE scores of these approaches were compared with those of the fine-tuned BART model to identify differences in summarisation performance.

The implemented workspace was examined against its intended functional requirements. This included logbook management, summary generation, supervisor feedback, real-time communication and integration with external services.

Functional verification considered whether individual features behaved as expected, while integration verification examined interactions between the frontend, backend and NLP microservice.

3.6 Ethics Privacy Security and Responsible AI

This research involves academic research logbooks collected from students, making data privacy and confidentiality important considerations. Appropriate data handling and access restrictions are essential to protect student information. The collaborative workspace incorporates authenticated access, including JSON Web Token (JWT) authentication for real-time communication. Furthermore, automated summaries and AI-assisted feedback suggestions may contain inaccuracies or omit important details. To address this, the fine-tuned BART model incorporates preprocessing and fallback mechanisms intended to reduce unreliable outputs. Supervisors can review generated summaries alongside original logbooks and modify AI-assisted feedback suggestions before providing feedback to students. Therefore, the system functions as an academic decision-support tool, ensuring that final supervisory decisions remain under human control.

Generative AI and automated tools were used to support the research and thesis preparation, as disclosed in Appendix B. ChatGPT was used for language refinement and content organisation, while Google Gemini was used for image generation. All AI-generated content was reviewed and verified by the author.

3.7 Reproducibility and Research Artefacts

The research can be reproduced using the collected academic research logbook dataset, reference summaries, fine-tuned BART model, preprocessing and evaluation scripts, and FastAPI microservice. The collaborative workspace can be rebuilt using the React frontend, Node.js backend and Supabase PostgreSQL database configuration. The source code and related development artefacts are maintained within the project repository, while access to student logbook data is subject to privacy restrictions.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The Collaborative Workspace & NLP-Powered Supervisor Assistant was designed to support academic research supervision through centralised logbook management, automated summarisation, communication and progress monitoring. The requirements were defined around the needs of two primary user roles: students and supervisors.

4.1.1 Functional requirements

Table 4.1: Functional Requirements of the Collaborative Workspace & NLP-Powered Supervisor Assistant

ID	Functional Requirement	Description	Verification Performed
FR01	Research logbook management	Allow students to submit and view research logbooks with automatic entry numbering and duplicate prevention.	Verified logbook creation, retrieval, chronological display and entry numbering.
FR02	Research logbook submission	Allow students to create and submit research logbooks containing activity records.	Verified successful logbook submission and storage in the database.
FR03	Logbook validation	Provide automatic entry numbering and prevent duplicate submissions for the same reporting period.	Verified automatic numbering and duplicate submission prevention.
FR04	Automated logbook summarisation	Generate concise summaries of submitted research logbooks using the fine-tuned BART model.	Verified summary generation through the FastAPI NLP microservice and display in the supervisor interface
FR05	Supervisor dashboard	Allow supervisors to view approved research groups, submitted logbooks and generated summaries.	Verified access to group information, submitted logbooks and generated summaries.
FR06	Supervisor review and feedback	Enable supervisors to examine original logbooks, provide feedback and approve or reject submissions.	Verified logbook review, feedback submission and approval or rejection functionality.
FR07	AI-assisted feedback	Generate contextual feedback suggestions using Google Gemini while	Verified AI suggestion generation, supervisor editing

	suggestions	allowing supervisors to review and modify them.	and feedback submission.
FR08	Real-time communication	Support communication between students and supervisors through Socket.IO-based group chat.	Verified real-time message delivery and display between authenticated users.
FR09	Research task management	Integrate Microsoft Planner to support research task organisation.	Verified Microsoft Planner integration and accessibility within the workspace.
FR10	Development activity monitoring	Integrate GitHub analytics to display repository activities and individual contributions.	Verified GitHub repository data retrieval and analytics display.
FR11	Announcements and notifications	Allow supervisors to publish announcements and provide relevant notifications to students.	Verified announcement creation, retrieval and display to intended users.
FR12	Logbook export	Support PDF export of research logbooks.	Verified PDF generation, logbook content and institutional formatting.

4.1.2 Non-functional requirements

The Collaborative Workspace & NLP-Powered Supervisor Assistant was designed to address the following non-functional requirements:

- **NFR1: Usability** - Provide user-friendly interfaces for students and supervisors to manage research activities efficiently.
- **NFR2 : Security** - Implement authentication and role-based access control to protect research information and user data.
- **NFR3: Reliability** - Ensure consistent system operation through appropriate error handling and service failure management.
- **NFR4: Data Integrity** - Maintain accurate logbook records through automatic entry numbering, and duplicate prevention.
- **NFR5: Responsible AI** - Allow supervisors to verify AI-generated summaries and review feedback suggestions before making final decisions.

These requirements guided the system architecture and implementation decisions discussed in the following sections.

4.2 Component Architecture Within the Complete Solution

The Collaborative Workspace & NLP-Powered Supervisor Assistant is an individual component of the ResearchLens platform. It follows a modular architecture consisting of a React frontend, Node.js and Express backend, Supabase PostgreSQL database and FastAPI-based NLP microservice. Figure 4.1 illustrates the architecture and interactions between these components.

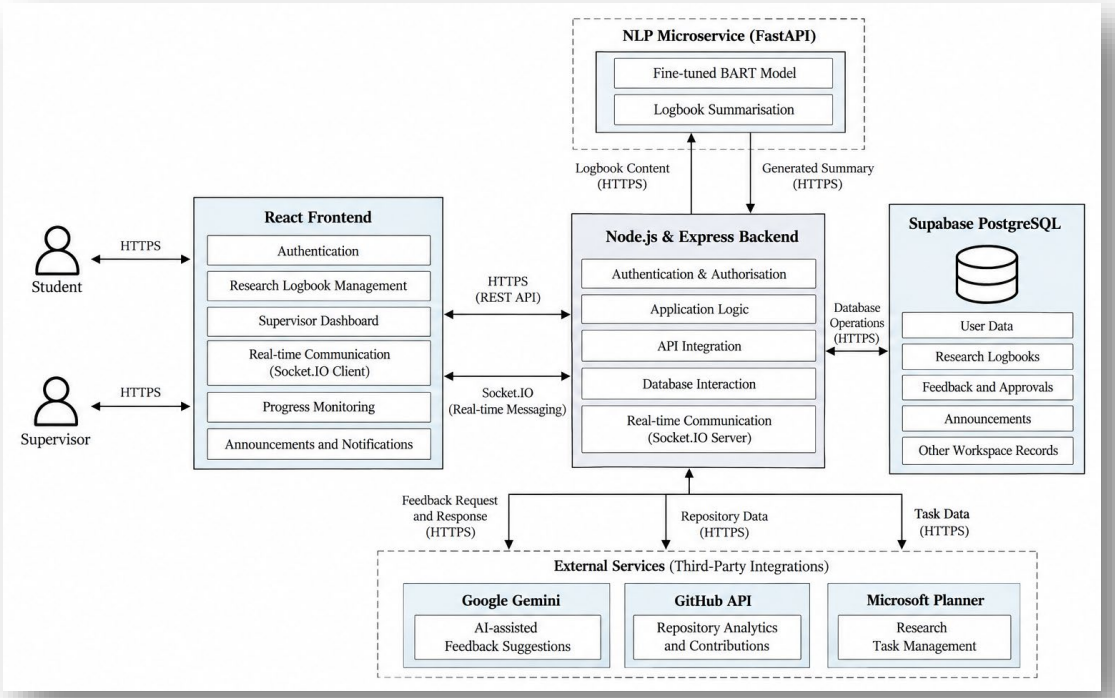


Figure 4.1: System Architecture of the Collaborative Workspace and NLP-Powered Supervisor Assistant

The architecture consists of the following components:

- **React Frontend:** Provides student and supervisor interfaces for research logbook management, progress monitoring, communication and feedback.
- **Node.js and Express Backend:** Handles application requests, authentication, database operations and communication with integrated services.
- **Supabase PostgreSQL Database:** Maintains user information, research logbooks, supervisor feedback and other workspace records.

- **NLP Microservice:** Uses FastAPI to expose the fine-tuned BART model for automated research logbook summarisation.
- **External Integrations:** Google Gemini supports AI-assisted feedback suggestions, Socket.IO enables real-time group communication, GitHub APIs provide repository analytics and Microsoft Planner supports task management.

When a student submits a research logbook, the backend processes and stores the submission. The logbook content is subsequently passed to the NLP microservice for summarisation, and the generated summary is returned to the workspace for supervisor review. Supervisors can examine the original logbook, review the generated summary and provide feedback, supported by editable AI-generated suggestions. The separation of frontend, backend, database and NLP processing supports maintainability and independent development of system components.

4.3 Design Alternatives and Decisions

Several design alternatives were considered when developing the Collaborative Workspace & NLP-Powered Supervisor Assistant. Table 4.3 summarises the principal design decisions.

Table 4.3: Summary of Design Alternatives and Selected Approaches

Design Decision	Alternatives Considered	Selected Approach	Reason for Selection
Summarisation model	BART, T5, PEGASUS and general-purpose LLMs	Fine-tuned BART	Supported domain-specific training and achieved the highest ROUGE scores among the evaluated models.
NLP service architecture	Direct backend integration; separate microservice	FastAPI microservice	Separates model inference from the main application backend.
Feedback generation	Manual feedback; AI-assisted suggestions	Google Gemini with supervisor review	Supports feedback preparation while retaining human oversight.

Real-time communication	Periodic message retrieval; WebSocket-based messaging	Socket.IO	Supports real-time group messaging between students and supervisors.
Research task management	Custom task management; external tool integration	Microsoft Planner	Provides task management capabilities without developing a separate planning system.
Development monitoring	Manual progress reporting; repository analytics	GitHub API integration	Provides repository activity information alongside research progress records.

The selected approaches establish a modular system in which research management, NLP processing and external integrations operate through defined interfaces. These decisions support the architecture presented in Section 4.2 and the detailed designs described in Section 4.4.

4.4 Detailed Design

4.4.1 Database design

The system uses Supabase PostgreSQL to manage student information, research groups, logbook submissions, feedback and announcements. These records support the relationships between students, supervisors and their assigned research groups. The database design also supports logbook retrieval and tracking of submission and review information.

4.4.2 NLP summarisation pipeline

The NLP summarisation pipeline processes submitted research logbooks using the fine-tuned BART model. Input text undergoes preprocessing and tokenisation before being passed to the model through the FastAPI microservice. The generated summary is then returned to the backend and displayed in the supervisor interface.

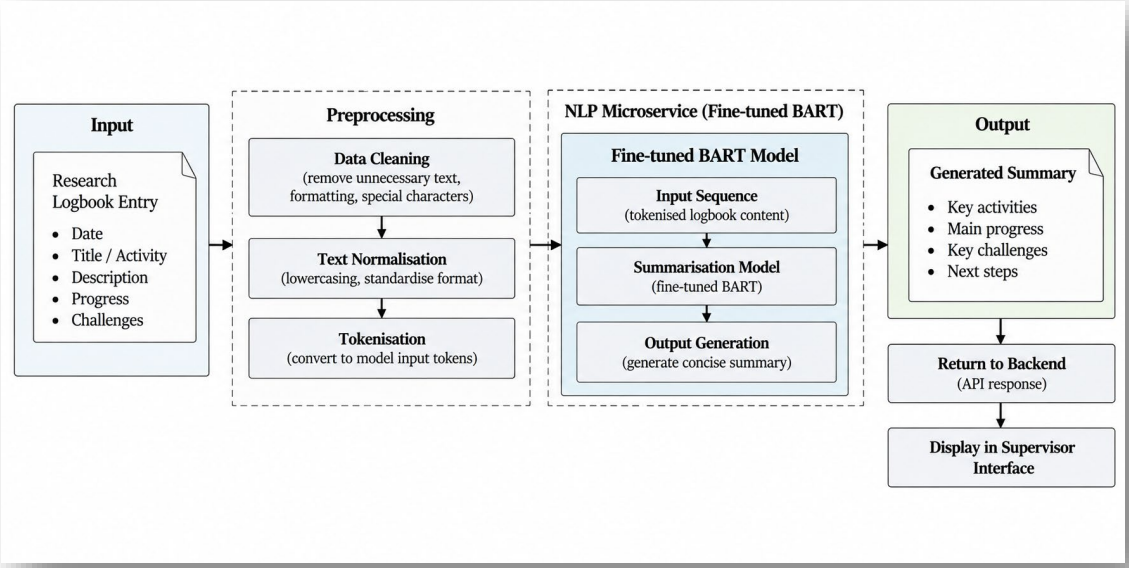


Figure 4.3: NLP-Based Research Logbook Summarisation Pipeline

Figure 4.3 illustrates the processing stages from logbook input to generated summary.

4.4.3 Student logbook interface

The student interface allows users to create and submit research logbooks containing descriptions of completed activities, progress and challenges. Submitted entries are maintained with automatic entry numbering and duplicate prevention.

The interface also supports viewing previously submitted logbooks and generating institutionally formatted documents for PDF export.

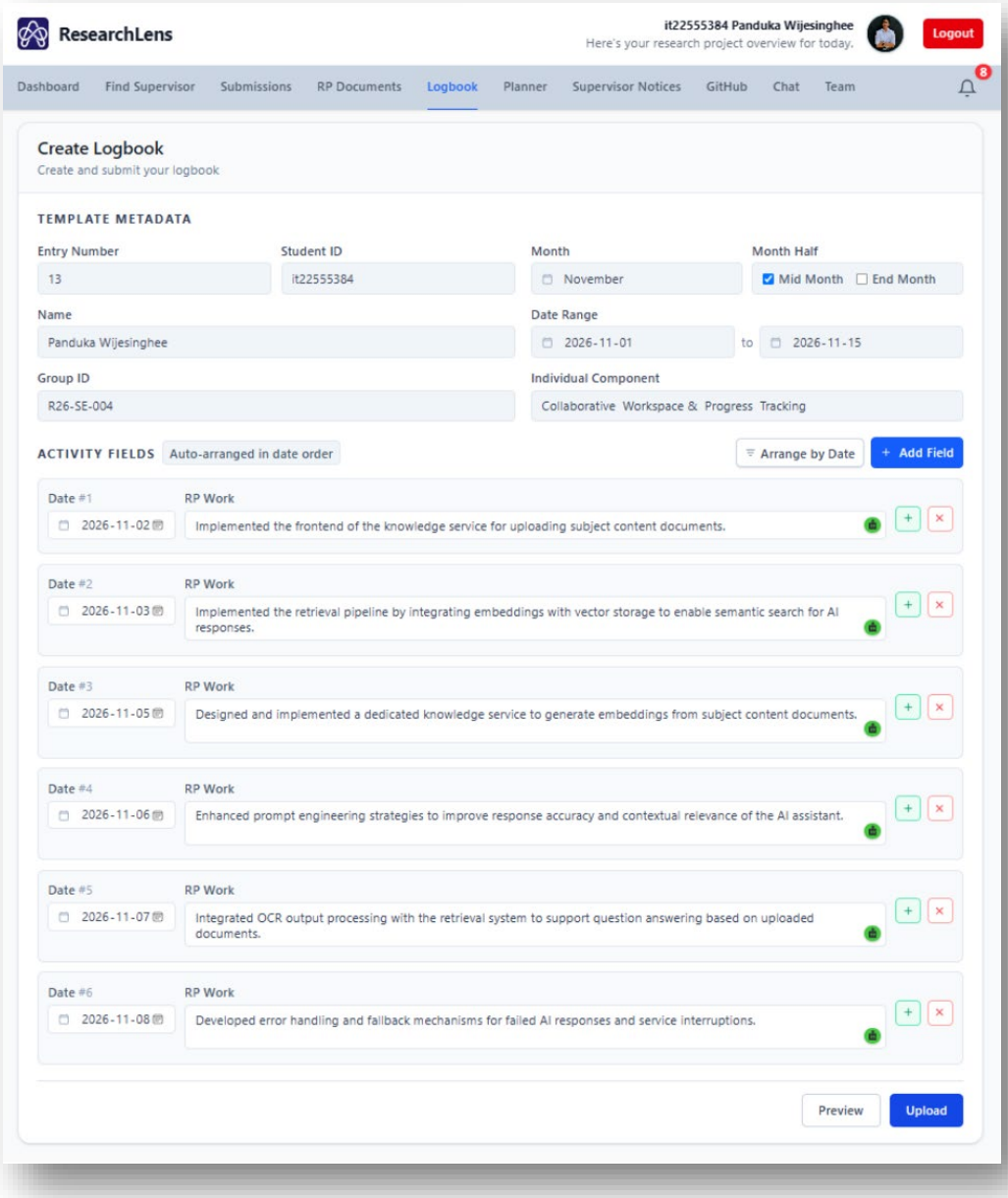


Figure 4.4: Student Logbook Submission Interface

4.4.4 Supervisor review and feedback interface

The supervisor interface provides access to assigned research groups, submitted logbooks and generated summaries. Supervisors can examine original entries alongside BART-generated summaries to understand reported research progress.

Google Gemini provides contextual feedback suggestions, which supervisors can review and modify before providing final feedback. This ensures that automated assistance supports rather than replaces academic judgement.

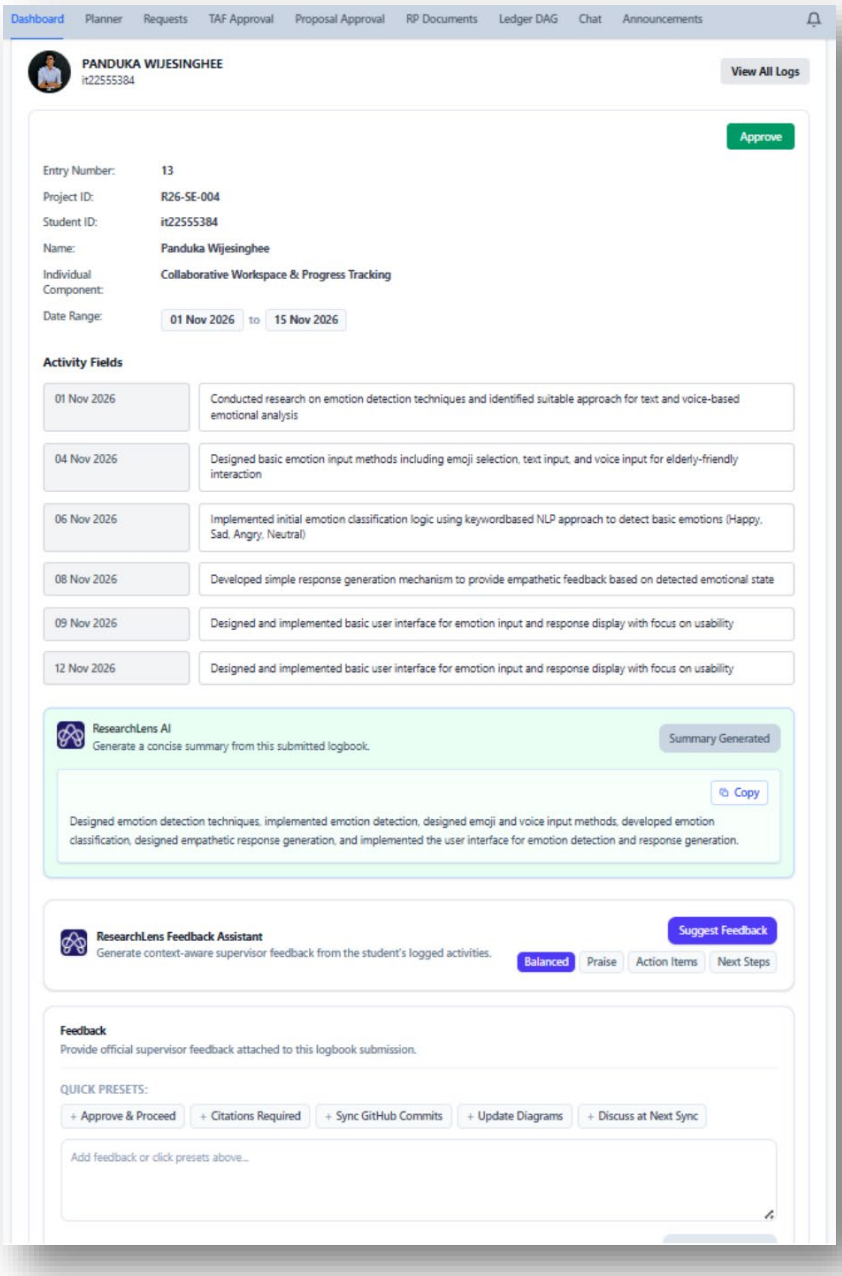


Figure 4.5: Supervisor Logbook Review Interface with Generated Summary

4.4.5 Communication and progress monitoring

The workspace uses Socket.IO to support real-time communication between research group members and supervisors. GitHub API integration provides repository activity information, while Microsoft Planner supports research task organisation. Announcements and notifications provide additional communication mechanisms within the workspace. These features complement logbook management by supporting research coordination and progress monitoring.

4.5 Standards and Professional Practice

The Collaborative Workspace & NLP-Powered Supervisor Assistant was designed following professional software engineering practices, with consideration for usability, security, maintainability and responsible AI. The modular architecture separates the frontend, backend and NLP microservice to support future enhancements, while authentication and access control help protect student research information. Logbook validation mechanisms maintain data integrity through automatic entry numbering and duplicate prevention. Furthermore, AI-generated summaries and feedback suggestions remain subject to supervisor review, ensuring human oversight in academic decision-making.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The Collaborative Workspace & NLP-Powered Supervisor Assistant was developed using a modular technology stack that supports research management, NLP processing and real-time communication. The React frontend communicates with the Node.js and Express backend, while the fine-tuned BART model operates through a separate FastAPI microservice.

Table 5.1: Technology Stack of the Collaborative Workspace & NLP-Powered Supervisor Assistant

Layer	Technology	Purpose
Frontend	React, JavaScript	Student and supervisor interfaces
Backend	Node.js, Express	Application logic and REST APIs
Database	Supabase PostgreSQL	Research logbooks, user information and feedback
NLP model	Fine-tuned BART	Automated research logbook summarisation
NLP microservice	Python, FastAPI	Model inference and API communication
Machine learning libraries	PyTorch, Hugging Face Transformers	Model fine-tuning and inference
AI feedback	Google Gemini API	Supervisor feedback suggestions
Real-time communication	Socket.IO	Group messaging
Development monitoring	GitHub API	Repository analytics
Task management	Microsoft Planner	Research task organisation
Frontend hosting	Netlify	Production frontend deployment
Backend and NLP hosting	Microsoft Azure Virtual Machines, Docker Compose	Containerised service deployment

5.2 Component Implementation

5.2.1 Research logbook management

The student interface allows users to create, submit and view research logbooks containing their activities, progress and challenges. Submitted entries are organised chronologically, with automatic entry numbering and validation to prevent duplicate submissions for the same reporting period. The system also supports generating institutionally formatted logbooks for PDF export.

5.2.2 BART summarisation model

The summarisation component was implemented using the pretrained facebook/bart-large-cnn model, fine-tuned on the academic research logbook dataset. PyTorch and Hugging Face Transformers were used for model development, while the trained model was integrated into a Python-based FastAPI microservice.

The microservice accepts logbook content, performs the required preprocessing and tokenisation, and generates a concise summary using the fine-tuned model. The generated output is returned to the application backend for presentation in the supervisor interface.

5.2.3 Supervisor review and AI-assisted feedback

The supervisor dashboard provides access to research groups, submitted logbooks and generated summaries. Supervisors can examine the original logbook alongside its summary and provide feedback on the student's reported progress.

Google Gemini was integrated to generate contextual feedback suggestions based on logbook content. These suggestions can be reviewed and edited by supervisors before submission, ensuring that the final feedback remains under human control.

5.2.4 Real-time communication

Real-time group communication was implemented using Socket.IO. The messaging component supports communication between students and supervisors, with JWT authentication used to verify access to protected messaging functions. Screenshots of the implemented group chat interface are provided in Appendix A.

5.2.5 Progress monitoring and supporting features

GitHub API integration retrieves repository activity information to support monitoring of development progress and individual contributions. Microsoft Planner is embedded within the workspace to provide access to research task management.

The implementation also includes announcements and notifications to support communication of research-related updates. Screenshots of the GitHub analytics dashboard, Microsoft Planner integration and announcements interface are presented in Appendix A.

5.3 Integration with the Other Components

The Collaborative Workspace & NLP-Powered Supervisor Assistant was integrated into ResearchLens, which consists of four research components: Recommendation and Collaboration, Similarity Analysis and Gap Detection, Integrity and Governance, and the Collaborative Workspace & NLP-Powered Supervisor Assistant. These components operate within a unified research management platform, supporting different stages of the academic research lifecycle.

The Recommendation and Collaboration component assists students in identifying suitable supervisors based on research interests, expertise and workload. Following supervisor selection and research group approval, the collaborative workspace supports ongoing supervision through research logbook submissions, progress monitoring and communication. The Similarity Analysis and Gap Detection component provides research novelty assessment by comparing proposed research topics against existing literature. The Integrity and Governance component focuses on maintaining research record integrity through cryptographic hashing, provenance tracking and blockchain-based verification.

Within this integrated environment, the individual component manages the student-supervisor workflow. Submitted research logbooks are stored in PostgreSQL and processed through the fine-tuned BART microservice. Generated summaries are displayed alongside original logbooks, allowing supervisors to review research progress and provide feedback. Shared application interfaces and backend services

provide a consistent user experience across ResearchLens, while independently deployed microservices maintain separation between the research components.

5.4 Verification and Software Testing

Table 5.2 summarises the principal verification cases based on the functional requirements identified in Section 4.1.

Table 5.2: Principal Test Cases and Verification Evidence

Test ID	Function Tested	Expected Result	Verification Evidence
T01	Logbook submission	Entries are submitted and stored successfully.	Submission and database records
T02	Logbook validation	Automatic numbering and duplicate prevention operate correctly.	Validation responses
T03	BART summarisation	The NLP microservice generates and returns a summary.	API response and supervisor interface
T04	Supervisor feedback	Supervisors can review logbooks and submit feedback.	Supervisor review interface
T05	AI-assisted feedback	Gemini suggestions can be generated and edited.	Feedback generation interface
T06	Real-time communication	Messages are delivered between authenticated users.	Group chat interface
T07	GitHub analytics	Repository activity information is retrieved and displayed.	GitHub analytics dashboard
T08	Microsoft Planner	The integrated planning interface is accessible.	Planner interface
T09	Announcements	Published announcements are visible to intended users.	Announcement interface
T10	Logbook export	Generated PDF documents contain correctly formatted logbooks.	Exported PDF document

5.5 Deployment and Operation

The ResearchLens platform React frontend is hosted on Netlify at https://researchlens.tech, with continuous integration and deployment (CI/CD) enabled through Git repository integration. The Node.js backend and FastAPI-based NLP microservice are deployed as Docker containers on Microsoft Azure Virtual Machines. Docker Compose manages the services independently, with the Workspace NLP microservice operating on container port 8010. Nginx Proxy Manager handles HTTPS routing, while Cloudflare provides an additional proxy and security layer.

6 RESULTS AND VALIDATION

6.1 Research Results

The research evaluated the summarisation performance of the fine-tuned BART model using a held-out test set of 10 academic research logbooks, representing 10% of the 100 logbook dataset. Performance was measured using ROUGE-1, ROUGE-2 and ROUGE-L to assess the similarity between generated summaries and reference summaries.

Table 6.1: Performance Comparison of Research Logbook Summarisation Models

Summarisation Model	ROUGE-1	ROUGE-2	ROUGE-L
Lead-3	0.3420	0.1125	0.2780
LexRank	0.3610	0.1230	0.2915
Pretrained BART	0.3710	0.1180	0.2950
ChatGPT (GPT-3.5-turbo)	0.3940	0.1315	0.3150
Fine-tuned T5-Base	0.4420	0.1625	0.3510
Fine-tuned PEGASUS	0.4615	0.1710	0.3680
Fine-tuned BART (Proposed Model)	**0.4785**	**0.1842**	**0.3810**

The fine-tuned BART model achieved a ROUGE-1 score of 0.4785, ROUGE-2 score of 0.1842 and ROUGE-L score of 0.3810. These results indicate that the generated summaries retained relevant vocabulary and textual sequences from the reference summaries. However, ROUGE measures lexical similarity and does not independently establish factual accuracy or summary usefulness.

The model was compared against six baseline approaches: Lead-3, LexRank, pretrained BART, fine-tuned T5, fine-tuned PEGASUS and ChatGPT. Among these, fine-tuned PEGASUS achieved the strongest baseline performance, with ROUGE-1, ROUGE-2 and ROUGE-L scores of 0.4615, 0.1710 and 0.3680, respectively. The fine-tuned BART model achieved higher scores across all three metrics.

These findings demonstrate that domain-specific BART fine-tuning produced better ROUGE results than the evaluated baseline approaches on the selected research logbook test set.

6.2 Product and Technical Results

The Collaborative Workspace & NLP-Powered Supervisor Assistant was successfully implemented and integrated into the ResearchLens platform. The developed component provides research logbook management, automated summarisation, supervisor feedback, real-time communication and research progress monitoring.

The student dashboard supports research logbook submission, entry management, automatic numbering and duplicate prevention. Submitted logbooks are stored in PostgreSQL and made accessible through the supervisor dashboard. The system also supports institutionally formatted logbook documents for PDF export.

The fine-tuned BART model was integrated through the FastAPI microservice, enabling automated summaries to be displayed alongside original logbooks. Google Gemini provides contextual feedback suggestions that supervisors can review and modify before submission, maintaining human oversight.

Additional implemented features include Socket.IO-based group communication, GitHub repository analytics, Microsoft Planner integration, announcements and notifications.

The technical results confirm that the principal components were implemented and connected within the ResearchLens platform.

6.3 User and Stakeholder Validation

The Collaborative Workspace & NLP-Powered Supervisor Assistant was designed to address the requirements of its two primary user groups: students and academic supervisors. Students can submit and manage research logbooks, monitor research progress through GitHub analytics, organise tasks using Microsoft Planner, communicate through real-time group chat, and view supervisor feedback, announcements and notifications. Academic supervisors can manage assigned

research groups, review submitted logbooks, access AI-generated summaries, provide feedback with AI-assisted suggestions, monitor development activities, communicate with students and publish announcements. These features support research coordination and supervision within a unified collaborative workspace.

6.4 Integration Results

The Collaborative Workspace & NLP-Powered Supervisor Assistant was integrated into the complete ResearchLens platform. The components operate within a shared application environment, supporting different stages of academic research management.

The individual component's end-to-end integration was demonstrated through student logbook submission, PostgreSQL storage, automated summarisation using the FastAPI-based BART microservice and supervisor review. Generated summaries were successfully displayed alongside original logbooks, while Google Gemini supported editable feedback suggestions. Real-time group messaging, GitHub analytics and Microsoft Planner were also integrated into the workspace.

The integration results establish that the collaborative workspace operates within the shared ResearchLens platform.

6.5 Summary of Findings

Table 6.2 summarises the principal research, product and integration findings of the Collaborative Workspace & NLP-Powered Supervisor Assistant, distinguishing experimentally evaluated results from implemented functionality.

Table 6.2: Summary of Research Findings and Supporting Evidence

Finding	Supporting Evidence	Validation Status
Fine-tuned BART outperformed six summarisation baselines across all three ROUGE metrics.	Held-out test dataset and Table 6.1	Quantitatively evaluated
Fine-tuned BART outperformed six summarisation baselines across all three ROUGE metrics.	Comparative model evaluation in Table 6.1	Quantitatively evaluated
Research logbook management,	Functional	Technically

automated summarisation and supervisor feedback were implemented.	verification and implemented interfaces	demonstrated
Real-time communication, GitHub analytics, Microsoft Planner and announcements were integrated.	Workspace implementation and integration evidence	Technically demonstrated
The collaborative workspace operates within the complete ResearchLens platform.	System integration and deployment evidence	Technically demonstrated

7 DISCUSSION

7.1 Interpretation of Findings

The findings demonstrate the potential of integrating domain-specific NLP summarisation into academic research supervision. The fine-tuned BART model achieved ROUGE-1, ROUGE-2 and ROUGE-L scores of 0.4785, 0.1842 and 0.3810, respectively, outperforming the six evaluated baseline approaches. These results address RQ1 by demonstrating improved lexical similarity between generated and reference summaries within the selected research logbook dataset.

Regarding RQ2, the implemented collaborative workspace provides students and supervisors with a centralised environment for logbook management, communication, feedback and progress monitoring. However, its impact on supervisory efficiency and research coordination was not quantitatively measured.

For RQ3, integrating BART-generated summaries and Google Gemini-based feedback suggestions into the supervisor workflow demonstrates the technical feasibility of AI-assisted supervision while maintaining human oversight. Supervisors retain access to original logbooks and control over final feedback.

Regarding RQ4, functional and integration verification supports the implementation of the intended system workflows, including logbook validation, summarisation, communication and external service integration.

7.2 Comparison with Literature Baselines and Existing Solutions

The research findings were compared with the summarisation approaches and academic supervision systems reviewed in Chapter 2. Table 7.1 highlights how the proposed component relates to existing techniques and identifies the evidence supporting its contribution.

Table 7.1: Comparison of Research Findings with Existing Literature

Area	Existing Research	This Research	Interpretation
Extractive summarisation	Lead-3 and LexRank select sentences from source documents.	Fine-tuned BART achieved higher ROUGE scores than both baselines.	Domain-specific abstractive summarisation performed better on the selected test dataset.
Transformer-based summarisation [4], [12], [14]	BART, T5 and PEGASUS support abstractive text generation.	Fine-tuned BART achieved the highest ROUGE scores among the evaluated models.	Fine-tuned BART was the strongest evaluated approach for research logbook summarisation.
Technology-supported supervision [1], [2], [8]	Digital approaches support communication, research coordination and feedback.	The workspace combines these functions with automated logbook summarisation.	Demonstrates technical integration of NLP into academic supervision.
Learning analytics and Git-based monitoring [3], [9]–[11]	Structured activity records provide indicators of student progress.	GitHub analytics complements narrative logbook summaries.	Supports multiple forms of progress information within one workspace.
AI in education [5]	LLMs offer opportunities for educational assistance but introduce reliability concerns.	Gemini-based feedback suggestions remain editable by supervisors.	Maintains human responsibility for academic feedback.

7.3 Research Contribution

The individual research contributes to AI-assisted academic supervision through the following outcomes:

- **Domain-specific research logbook summarisation:** A fine-tuned BART model adapted to academic research logbooks, achieving higher ROUGE scores than six evaluated summarisation baselines.

- **Integration of NLP into academic supervision:** A framework that connects automated logbook summarisation with student submissions and supervisor review, while maintaining human oversight.
- **Combined research progress representation:** An approach that brings together narrative research logbooks, generated summaries and GitHub development activity information to support supervision.

7.4 Product Contribution

The individual product contribution is the development and integration of the Collaborative Workspace & NLP-Powered Supervisor Assistant within the ResearchLens platform. The implemented component provides a centralised environment for students and academic supervisors to manage research activities, communication and feedback.

The product includes research logbook submission and management with automatic entry numbering, duplicate prevention and PDF export. A fine-tuned BART microservice generates summaries for supervisor review, while Google Gemini provides editable feedback suggestions. Additional features include Socket.IO-based real-time group communication, GitHub repository analytics, Microsoft Planner integration, announcements and notifications.

7.5 Limitations and Threats to Validity

- **Dataset size:** The BART model was fine-tuned using 100 academic research logbooks, with only 10 samples reserved for testing. This limits the generalisability of the results to other research domains and institutions.
- **Evaluation limitations:** ROUGE metrics measure lexical similarity but do not directly assess factual accuracy, readability or completeness.
- **Technical limitations:** Large-scale performance, concurrent user handling and system scalability were not quantitatively evaluated. The system also depends on external services, including Google Gemini, GitHub and Microsoft Planner.

7.6 Implications

The research findings have implications for the development of AI-assisted academic supervision systems. The results suggest that domain-specific abstractive summarisation can support the processing of research logbooks by presenting condensed information for supervisor review. However, generated summaries should complement rather than replace the examination of original student submissions.

From a practical perspective, the Collaborative Workspace & NLP-Powered Supervisor Assistant demonstrates how research documentation, communication, progress monitoring and AI-assisted feedback can be combined within a single platform. This approach provides a foundation for developing more integrated academic supervision tools.

The findings also highlight the importance of maintaining human oversight when applying generative AI to academic feedback.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research investigated the application of natural language processing and collaborative technologies to support academic research supervision through the development of the Collaborative Workspace & NLP-Powered Supervisor Assistant, integrated within the ResearchLens platform. The study focused on automating research logbook summarisation while providing a centralised environment for research documentation, progress monitoring, communication and supervisor feedback.

A fine-tuned BART model was developed and evaluated using 100 academic research logbooks. The model achieved ROUGE-1, ROUGE-2 and ROUGE-L scores of 0.4785, 0.1842 and 0.3810, respectively, outperforming six evaluated summarisation baselines. These findings demonstrate the potential of domain-specific fine-tuning for generating concise academic research progress summaries.

The implemented workspace integrates automated summarisation, AI-assisted feedback suggestions, logbook management, real-time communication, GitHub analytics, Microsoft Planner and supervisor announcements. The integration demonstrates the technical feasibility of combining NLP-based assistance with collaborative research supervision while maintaining human control over academic feedback. However, the research was limited by its relatively small dataset.

8.2 Achievement of Objectives

Objective O1: Collaborative Academic Research Supervision Workspace

This objective was achieved through the implementation of a centralised workspace supporting student logbook submissions, supervisor review, research progress monitoring and feedback management. Separate student and supervisor interfaces were developed to support their respective research supervision activities.

Objective O2: NLP Summarisation and AI-Assisted Feedback

This objective was achieved through the integration of a fine-tuned BART summarisation service and Google Gemini-based feedback suggestions into the

supervisor review workflow. The BART model outperformed six evaluated baselines across all three ROUGE metrics. Supervisors retain access to original logbooks and control over final feedback, ensuring human oversight.

Objective O3: Research Collaboration and Progress Monitoring

This objective was addressed by implementing Socket.IO-based real-time group communication, GitHub repository analytics, Microsoft Planner integration, supervisor announcements and notifications. These features provide students and supervisors with access to communication, task management and development progress information within the collaborative workspace.

Objective O4: Research Logbook Management and Verification

This objective was addressed through the implementation and functional verification of research logbook management features, including automatic entry numbering, duplicate submission prevention, chronological activity management and institutional print formatting. These functions support consistent research documentation and supervisor review.

8.3 Individual Research and Product Contributions

The individual contribution to ResearchLens consists of two main areas: the research and evaluation of NLP-based academic logbook summarisation and the development of the Collaborative Workspace & NLP-Powered Supervisor Assistant.

8.3.1 Research Contribution

The research involved preparing a dataset of 100 academic research logbooks, fine-tuning a BART model and evaluating its summarisation performance against six baseline approaches. The results demonstrated improved ROUGE scores for the fine-tuned model, providing experimental evidence for domain-specific abstractive summarisation in academic research supervision.

8.3.2 Product Contribution

The developed component includes student and supervisor dashboards, research logbook management, automated summarisation, AI-assisted feedback suggestions, real-time group communication, GitHub repository analytics, Microsoft Planner

integration, announcements and notifications. The implementation also involved integrating the React frontend, Node.js backend, PostgreSQL database and FastAPI-based NLP service into the shared ResearchLens platform.

8.4 Future Directions

Although the Collaborative Workspace & NLP-Powered Supervisor Assistant achieved its primary research and implementation objectives, several opportunities remain for further development and evaluation.

- **Dataset Expansion:** Expand the academic research logbook dataset to include different research disciplines, institutions and writing styles to improve the generalisability of the summarisation model.
- **Enhanced Summarisation Evaluation:** Conduct human evaluations involving academic supervisors to assess the factual accuracy, readability, completeness and usefulness of generated summaries beyond ROUGE metrics.
- **Scalability and Performance Testing:** Evaluate system performance under increased user activity, concurrent logbook submissions and real-time communication workloads to assess its suitability for wider institutional deployment.

These improvements would strengthen the reliability, usability and scalability of the proposed component and support its potential adoption in academic research supervision environments.

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GLOSSARY

Term	Definition
Academic Research Logbook	A structured document used by students to record research activities, progress, challenges and achievements.
AI-Assisted Feedback	Feedback suggestions generated using artificial intelligence to support supervisors in reviewing student research activities.
Collaborative Workspace	A centralised digital environment that enables students and supervisors to manage research activities, communicate and exchange feedback.
Fine-Tuning	The process of adapting a pretrained machine learning model to a specific task using a relevant dataset.
GitHub Analytics	The analysis and visualisation of repository activities, such as commits and development contributions, to support research progress monitoring.
Microservice	An independently deployable software service that performs a specific function within a larger application.
Pretrained Model	A machine learning model that has been trained on a large dataset and can be adapted to perform specific tasks.
Supervisor Assistant	A software component that supports academic supervisors through automated logbook summarisation, AI-assisted feedback and research progress monitoring.

APPENDICES

Appendix A Research Instruments and Extended Results

Implementation Screenshots

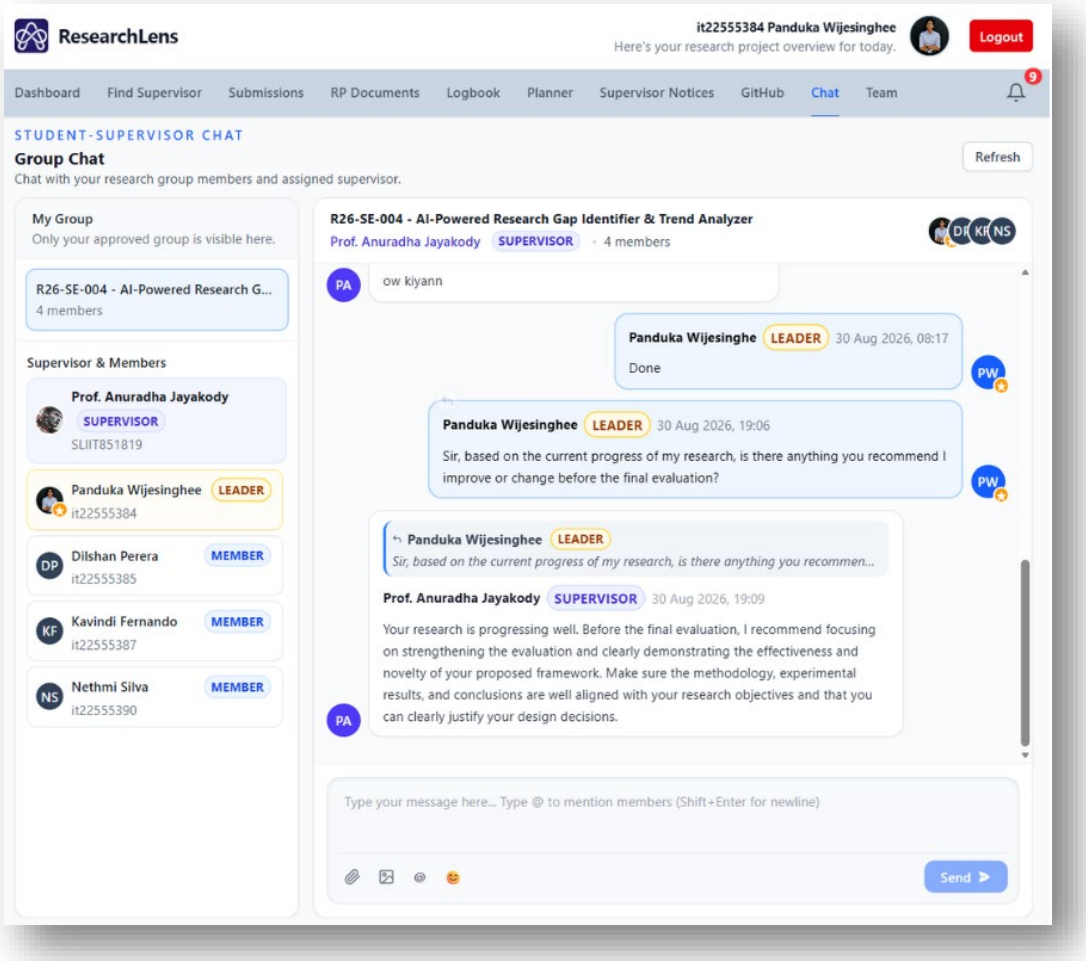


Figure A.1: Real-Time Group Chat Interface of the Collaborative Workspace

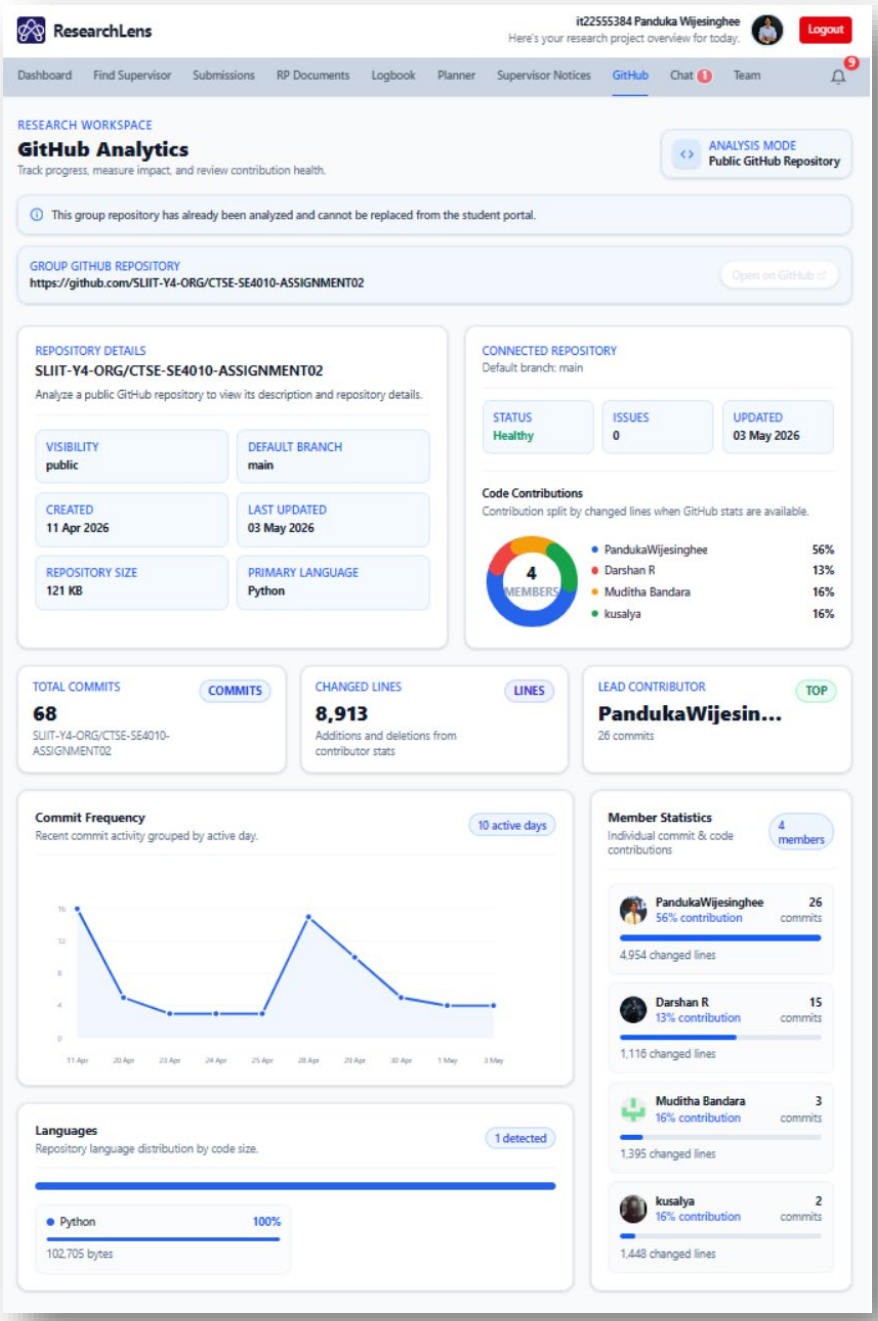


Figure A.2: GitHub Repository Analytics Dashboard

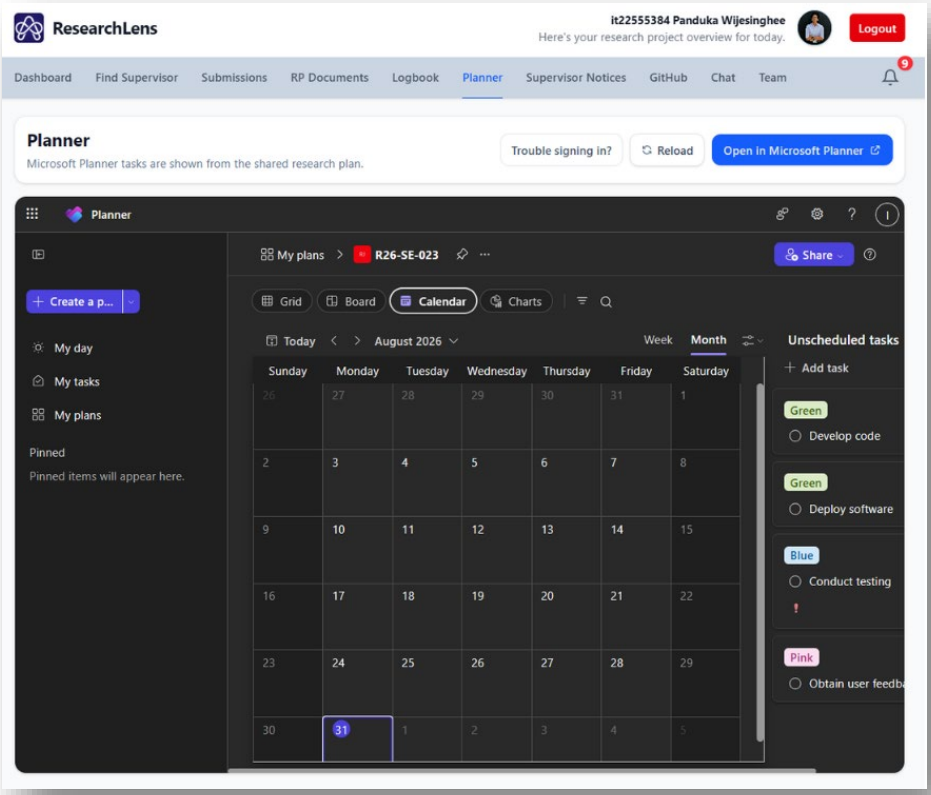


Figure A.3: Microsoft Planner Integration for Research Task Management

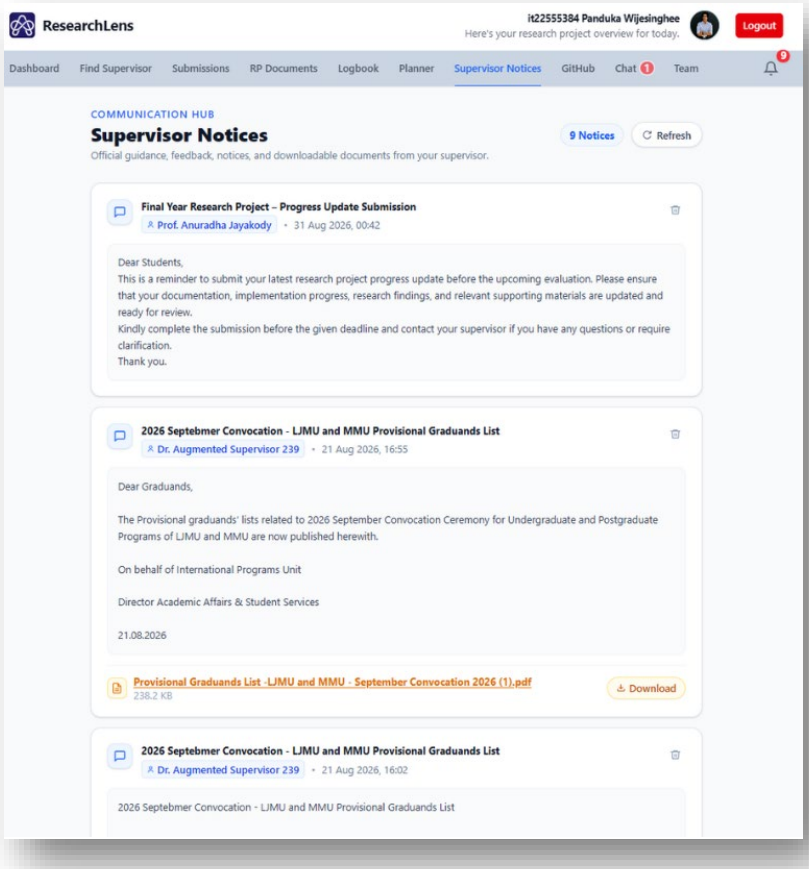


Figure A.4: Supervisor Notifications Interface of the Collaborative Workspace

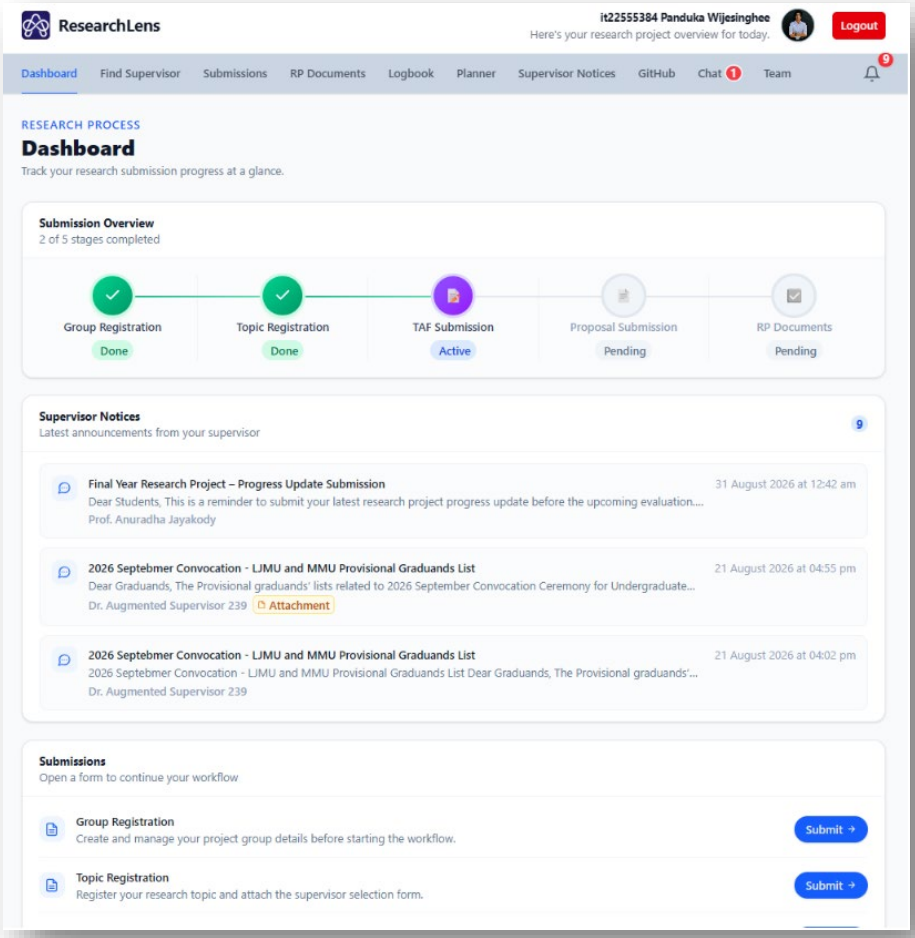


Figure A.5: Student Dashboard of the Collaborative Workspace

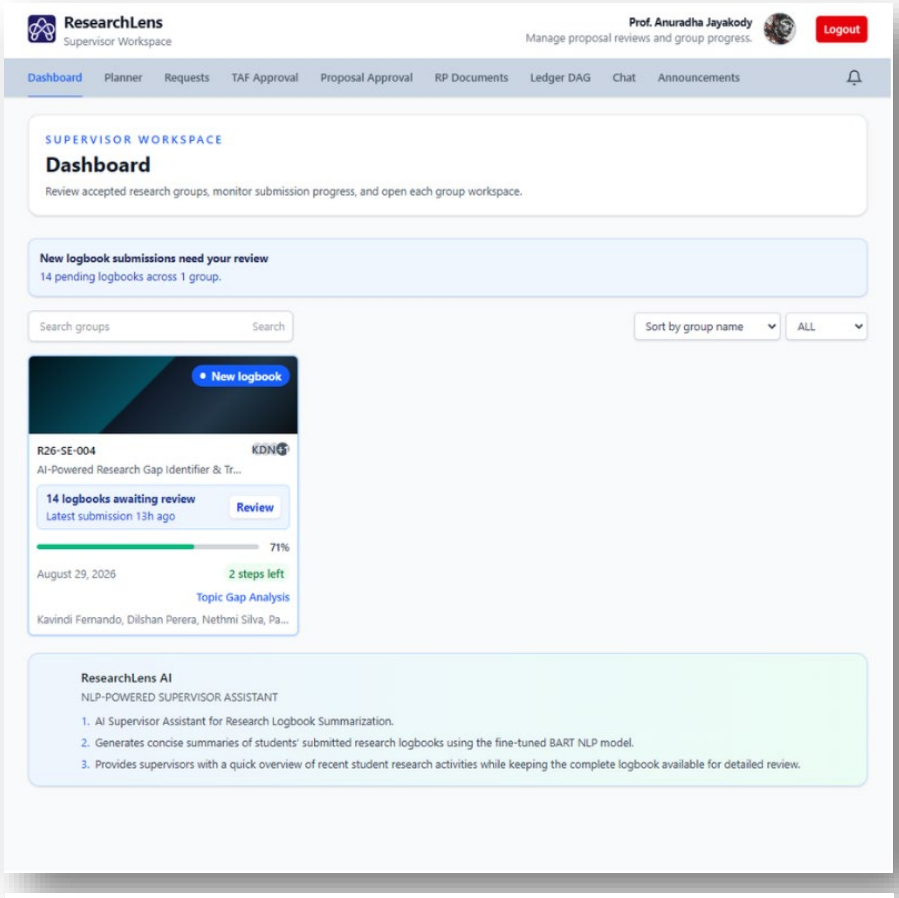


Figure A.6: Supervisor Dashboard of the Collaborative Workspace

Appendix B Academic Integrity and AI Use Disclosure

Artificial intelligence tools were used during the development and documentation of the Collaborative Workspace & NLP-Powered Supervisor Assistant. Their purposes and the extent of their involvement are outlined below.

Table B.1: AI Tools and Their Purposes in Research Development and Thesis Preparation

AI Tool	Purpose of Use
BART	Fine-tuned and evaluated as the primary NLP model for automated academic research logbook summarisation.
Google Gemini	Integrated into the collaborative workspace to generate editable feedback suggestions for academic supervisors.
ChatGPT	Assisted with academic writing, language refinement, content organisation and technical explanations during thesis preparation.

The fine-tuned BART model and Google Gemini integration form part of the research implementation, while ChatGPT was used as a supporting tool during thesis preparation.

AI-generated writing suggestions should be reviewed for accuracy, relevance and consistency with the research findings. The researcher remains responsible for the final thesis content, implementation decisions, experimental results and conclusions.